Practice workbook · Explain sex-linked inheritance and why males are affected more

X-linked recessive conditions such as haemophilia and colour blindness, contrasted with Y-linked traits.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-SEX-LINKED-INHERITANCE

1. Question 7 · part b · 4 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b only.

7 The four blood groups that occur in humans are group A, group B, group AB and group O.

There are three blood group alleles: I^A, I^B and I^O.

I^O is recessive to both I^A and I^B.

I^A and I^B are co-dominant to each other.

(a) (i) Explain what is meant by the term **co-dominant**.

(2)

(ii) State the possible genotypes of a person with blood group B.

(2)

(b) Haemophilia is a sex-linked condition that prevents the blood clotting.

Explain why males are more likely to be affected by haemophilia than females.

(4)

(Total for Question 7 = 8 marks)

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1. Question 7 · part b · 4 marks · Working and answer

Working and answer

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2. Question 6 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

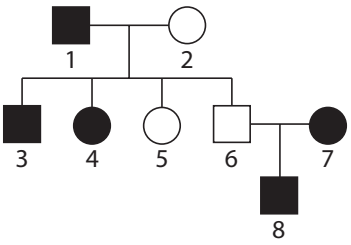
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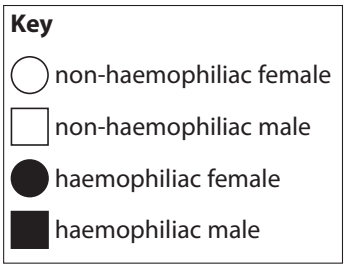
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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2. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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3. Question 6 · part b.iv · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

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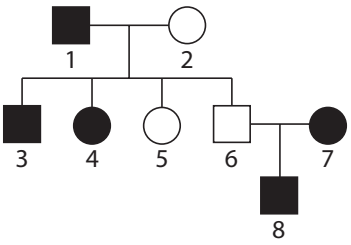
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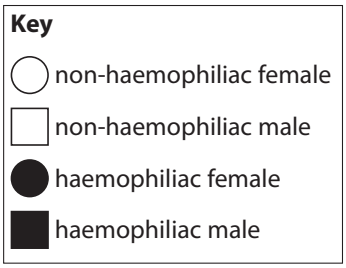
(ii) State what is meant by the term **sex-linked**.

(2)

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- ☒ **C** $X^H Y$
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3. Question 6 · part b.iv · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

(ii) What is the genotype of person 5?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

probability = ...

(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

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(Total for Question 6 = 12 marks)

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3. Question 6 · part b.iv · 3 marks · Working and answer

Working and answer

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4. Question 8 · part c · 5 marks

November 2020 Paper 2

Further practice · Attempt part c only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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4. Question 8 · part c · 5 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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(ii) The man and woman have a second child.

Calculate the probability that both children will have haemophilia.

(3)

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(c) Explain why the man cannot be a carrier of haemophilia.

(5)

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(Total for Question 8 = 15 marks)

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4. Question 8 · part c · 5 marks · Working and answer

Working and answer

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5. Question 9 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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5. Question 9 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

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(b) Haemophilia is a sex-linked genetic condition that affects the blood.

(i) Describe what is meant by a sex-linked genetic condition.

(2)

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(ii) State one effect that haemophilia will have on an individual.

(1)

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.....

.....

(Total for Question 9 = 10 marks)

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Working and answer

6. Question 5 · part b.i · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.i only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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6. Question 5 · part b.i · 2 marks

November 2024 Paper 2

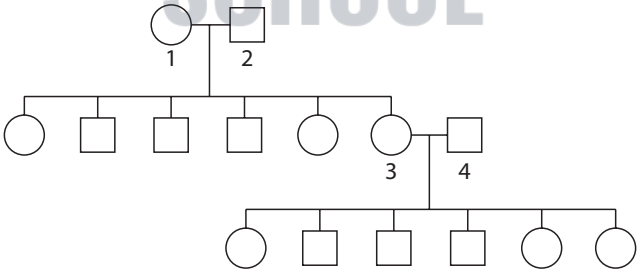
Further practice · Attempt part b.i only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

(2)

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.



Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

(2)

person 1

person 2

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6. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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7. Question 6 · part a.iii · 5 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

7. Question 6 · part a.iii · 5 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

(iii) Justify the statement that colour blindness is a recessive sex-linked condition.

(5)

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7. Question 6 · part a.iii · 5 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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(b) A woman who is heterozygous for the red-green colour blind condition has a child. The father does not have the condition.

Use a genetic diagram to determine the possible genotypes and phenotypes of the child.

Use B to represent the allele for normal vision and b to represent the allele for the colour blind condition.

(4)

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(Total for Question 6 = 13 marks)

7. Question 6 · part a.iii · 5 marks · Working and answer

Working and answer

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